

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE

02

Light – Part - 01



Light – Part - 01

Colour Intermixing in light

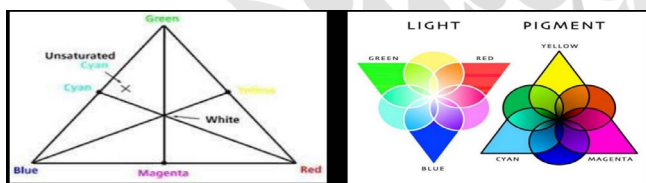
In light, colour intermixing follows additive mixing, where different colored lights combine to produce new colors.

The primary colors are :—

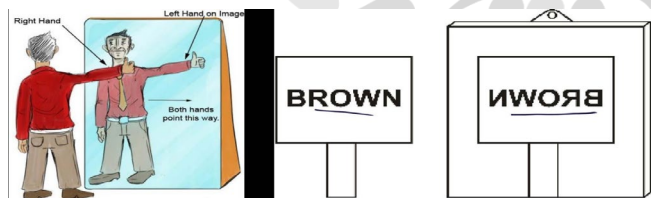
- Red (R), Green (G), Blue (B)

Rules of Additive Colour Mixing

- Red + Green = Yellow
- Red + Blue = Magenta
- Green + Blue = Cyan
- Red + Green + Blue = White light



Reflection of Light



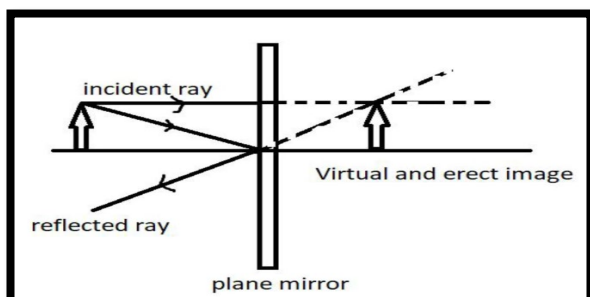
Mirror

A mirror is a smooth, highly polished surface that reflects light according to the laws of reflection.

Types of Mirrors

Plane Mirror

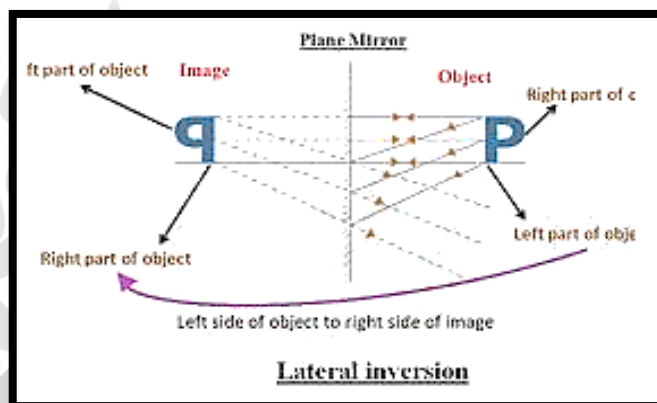
- Image is virtual, erect, same size
- Image distance = object distance
- Lateral inversion



If object moving with 'U' velocity towards mirror than find out the velocity of image with respect to object.

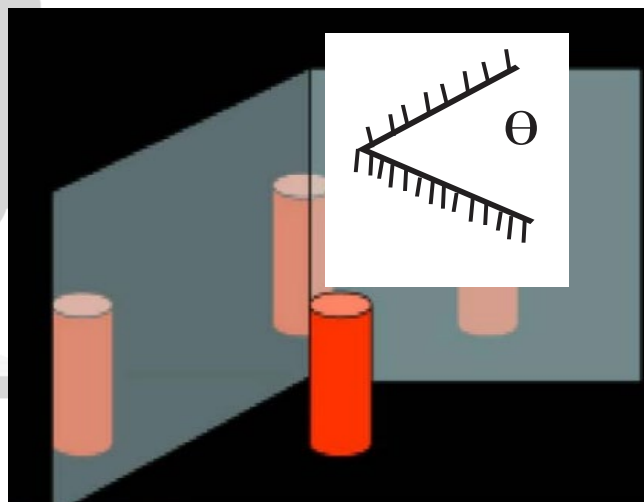
Velocity of image — U
Velocity of image w.r.t. object = $U + U = 2U$ m/s

Lateral Inversion in Plain Mirror



Formation of images by two plane mirrors

An object is present between two plane mirrors, which is kept at certain angle, number of images formed



Case -I

$$(i) \frac{360^\circ}{\theta} = m \rightarrow \text{odd no.}$$

Number of images = m

Case -II

$$\frac{360^\circ}{\theta} = m \rightarrow \text{even no. (2,4,6,8 ...)}$$

$$N = m - 1$$

∴ no of images



Case – III

If object is preset at the bisector position of two inclined mirror.

$$\frac{360^\circ}{\theta} = m$$

No. of images = $n = m - 1$

$$\frac{360^\circ}{\theta} = n = 6 - 1 = 5$$

Case – IV

If value of $\left(\frac{360^\circ}{\theta} = m\right)$ comes in decimal.

e.g. — $m = 3.8$ [$n = 3$]

Format $m = n. xyz$

If $n = \text{odd number}$, no. of images will be = n

If $n = \text{even number}$, no. of images will be = $n - 1$

$m = 4.8$ [$n = 4.1$] = 3

[$m = 2.3$ $n = 2 - 1 = 1$]

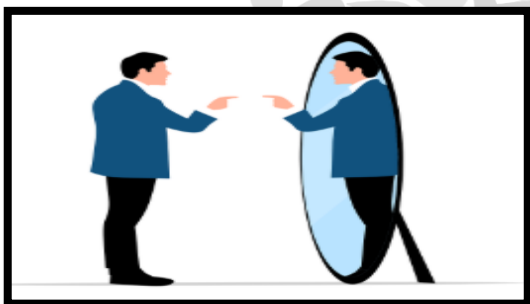
Case- V

When mirrors are parallel ($\theta = 0^\circ$)

Infinite images formed = infinity

Due to continuous multiple reflections

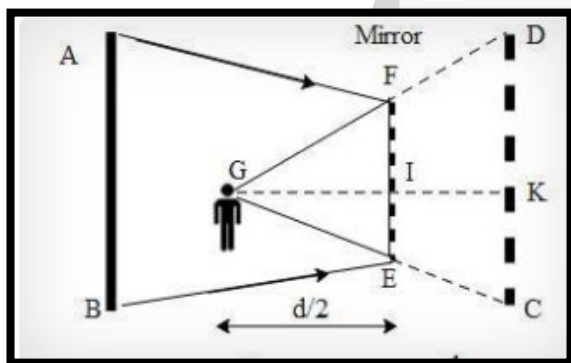
Minimum height of plane mirror required to see full image itself:



Minimum length of plane mirror (l) = $H/2$

H = Height of the person

Minimum height of plane mirror required to see full image wall behind itself



Minimum length required of plane mirror = $H/3$

$L = 15/3 = 5 \text{ cm}$

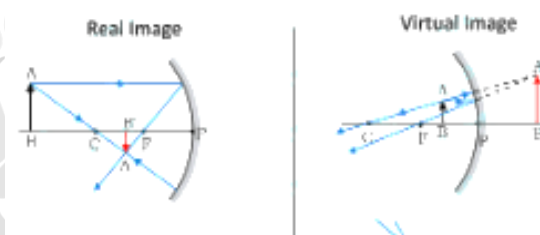
H = Height of behind wall or building.

Spherical Mirrors

Spherical mirrors are mirrors whose reflecting surface is a part of a sphere.

Concave mirror

- Forms real & inverted image (except when object is very close)
- Used in shaving mirrors, telescopes, headlights
- Curved mirror is a part of a hollow sphere. If reflection takes place from the inner surface, then the mirror is called concave and if its outer surface acts as reflector it is convex.



Concave mirror	Convex mirror
• Part of Hollow sphere whose inside surface is reflecting surface. • Converging in nature. • Generally, forms real and inverted image.	• Part of Hollow sphere whose outside surface is reflecting surface. • Diverging in nature. • Always, forms virtual and erect image.

DEFINITIONS FOR THIN SPHERICAL MIRRORS

- **Pole:** Pole is any point on the reflecting surface of the mirror. For convenience we take it as the central P of the mirror
- **Principal:** any section of the mirror such as MM' passing through the pole.
- **Centre of curvature** is Centre C of the sphere of which the mirror is a part.
- **Radius of curvature** is the radius R of the sphere of which the mirror is a part.
- **Principal axis** is the line CP, joining the pole and Centre of curvature of the mirror.
- **Principal focus** is an image point F on the principal axis for which object is at object infinity.
- **Focal length** is the distance PF between pole P and focus F along the principal. $F = R/2$.



- **Linear Aperture**, in reference to a mirror, is the effective diameter of the light reflecting area of the mirror.
- **Focal Plane** is the plane passing through focus and perpendicular to the principal axis.

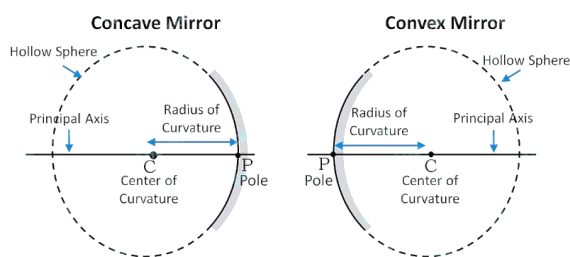
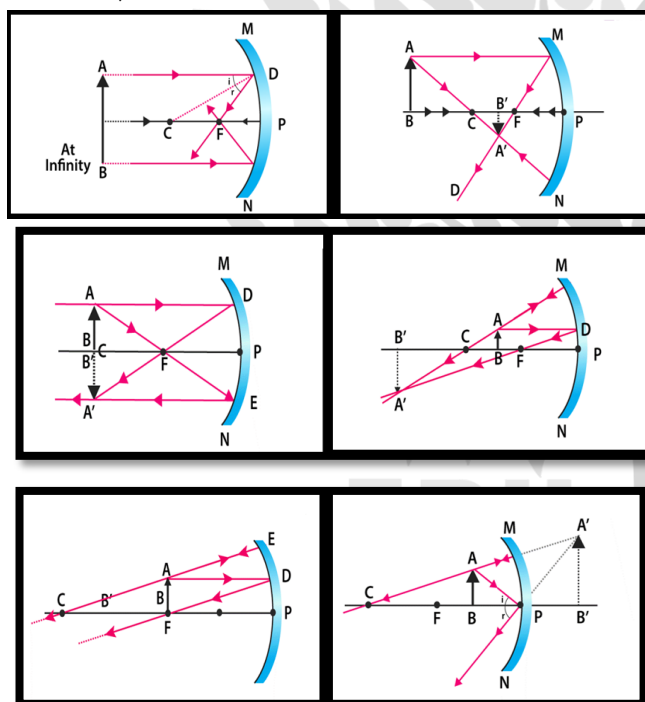


Image formation in spherical mirror

1. Concave Mirror

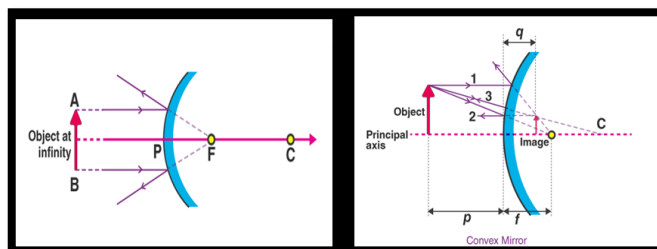
- Real & inverted image (object beyond focus)
- Virtual & erect image (object between pole and focus)



Sl. No.	Position of Object	Position of Image	Nature of Image	Size of Image
1	At infinity	At focus (F)	Real & inverted	Highly diminished (point-sized)
2	Beyond Centre of curvature (Beyond C)	Between C and F	Real & inverted	Diminished
3	At Centre of curvature (At C)	At C	Real & inverted	Same size
4	Between C and F	Beyond C	Real & inverted	Magnified
5	At focus (At F)	At infinity	Real & inverted	Infinitely large
6	Between F and pole (P)	Behind the mirror	Virtual & erect	Magnified

2. Convex Mirror

- Virtual, erect, diminished image



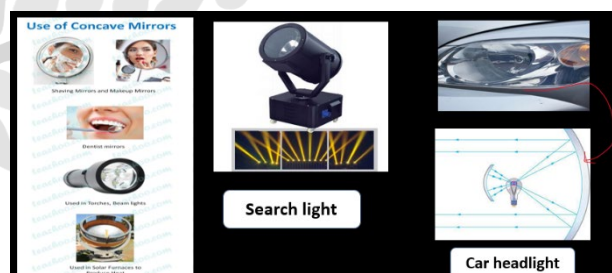
Uses of Mirror

I. Uses of Plane Mirror

- Looking/dressing mirror → forms same size image
- Periscope → used in submarines, tanks
- Kaleidoscope → multiple reflection patterns
- Optical instruments (e.g., reflecting devices)

II. Uses of Concave Mirror

- Shaving/makeup mirror → enlarged image (object close)
- Dentist mirror → magnified view of teeth
- vparallel beams
- Solar cooker/furnace → concentrates sunlight at focus



III. Uses of Convex Mirror

- Rear-view mirrors (vehicles) → wide field of view
- Security/surveillance mirrors → shops, parking areas
- Road safety mirrors → at blind turns.

